CPU scheduling

CPU and I/O bursts provide a model for comparing fairness, throughput, response, priorities, deadlines, and multiprocessor costs. A useful scheduling claim specifies both its workload and the guarantees it promises.

Prerequisites. Process states and queues, Context switching, Probability and expectation.

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Learning objectives

- Derive waiting, turnaround, and response times
- Compare FCFS, SJF, SRTF, RR, priority scheduling, and feedback queues
- Analyze preemption, starvation, priority inversion, and real-time schedulability

Central question. Under which workload and execution assumptions can a policy allocate finite CPU capacity correctly and predictably?

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6.1. Basic concepts

- A process alternates between CPU execution and I/O waits. A CPU burst is computation between blocking events; scheduling may interrupt that burst without completing its CPU demand.
- The short-term scheduler selects a runnable kernel execution context. The dispatcher saves and restores context, performs required mode transitions, and resumes execution. Dispatch latency is overhead rather than application progress.

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6.1.1. The scheduler's time scales

- Long-term admission controls which jobs enter and affects multiprogramming. Short-term scheduling selects the next execution context. Classical medium-term scheduling suspends or swaps work to reduce memory pressure.
- Admission changes the future workload. Batch throughput and interactive response therefore cannot be optimized by treating every decision as one CPU-selection policy.

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6.1.2. Preemptive and nonpreemptive policies

- A nonpreemptive policy switches when the running task blocks, terminates, or yields. A preemptive policy can also switch on a quantum expiration or a newly ready higher-priority task.
- Preemption can improve responsiveness but adds state and run-queue work and permits more interleavings of shared kernel data.

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6.1. Model, ordering, and boundary cases

Model assumptions. Jobs have defined arrivals and CPU/I/O demands. Context can be saved and restored, and queue transitions are synchronized.

Invariant and ordering. Each modeled CPU runs at most one task. A task cannot be enqueued twice, run before arrival, or use an unreleased resource.

Counterexample or boundary. Ignoring switching cost can make an extremely short quantum look fair while useful CPU time collapses.

Worked check. With P1 burst 5 before I/O, P2 burst 2, one CPU and switch cost 1, FCFS runs P1[0,5), switch[5,6), P2[6,8). Useful-work fraction is 7/8=87.5% and switching overhead is 1/8=12.5%. The CPU is never idle, so busy-time CPU utilization is still 8/8=100%. These are different measures; omitting switching also incorrectly changes elapsed time from 8 to 7.

Reasoning prompt. Which event-order bug could both duplicate a queue entry and permit execution in the wrong state?

6.2. Scheduling criteria

- Utilization and throughput generally favor more completed useful work. Turnaround, waiting, and response favor lower delays. These objectives can conflict.
- Waiting time accumulates only in the ready state, excluding CPU service and blocked I/O. Response must be defined as first dispatch or first externally visible output; the two need not coincide.

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6.2.1. Formulas and observation intervals

- Let arrival be a_i, first dispatch s_i, and completion f_i. Turnaround_i=f_i-a_i. Under the first-dispatch convention, response_i=s_i-a_i. Waiting_i is the sum of ready intervals, not generally turnaround minus CPU time when blocked I/O exists.
- A window that omits unfinished long jobs systematically favors short completions. Report the treatment of censored jobs rather than silently dropping them.

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6.2. Model, ordering, and boundary cases

Model assumptions. All jobs use the same time units, arrival definition, and completion criterion. The observation includes jobs still outstanding at its endpoint.

Invariant and ordering. Turnaround equals completion minus arrival. With compatible definitions, response does not exceed turnaround. Ready waiting excludes blocked time.

Counterexample or boundary. Maximizing utilization by indefinitely deferring short interactive work can violate response requirements.

Worked check. Arrival a=2, first CPU s=5, first visible output o=9, completion f=14, with [2,5) in the ready queue and no other ready waiting. Waiting=3, scheduling response=3, visible response=7, turnaround=12. They describe overlapping intervals and cannot be added.

Reasoning prompt. How can equal mean waiting coexist with very different maximum response?

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Ready waiting and two response events

Let arrival be a_i, total ready intervals W_i, first CPU s_i, first visible output o_i, and completion f_i. Then

$$waiting_i = W_i, \quad turnaround_i = f_i - a_i, \quad schedule\ response_i = s_i - a_i, \quad visible\ response_i = o_i - a_i.$$

Blocked I/O is neither service nor ready waiting. Task A arrives at 0, is ready[0,2), runs[2,4), blocks[4,8), is ready[8,11), then runs[11,12). Turnaround=12, ready waiting=2+3=5, service=2+1=3, blocked=4. The decomposition 5+3+4=12 closes. Counting blocked time as waiting gives 9 and violates the definition.

Equal mean waiting need not imply equal maximum response. A and B can each accumulate 4 ready units, yet first run at 4 and 8 under different arrivals/blocking. First visible output can occur still later. Metric names, observed events, units, and percentiles must accompany comparisons.

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6.3. Scheduling algorithms

- Define the comparison key, preemption rule, tie-break, and queue update. A lower mean wait does not establish a per-job bound.
- Nonpreemptive SJF minimizes mean wait for simultaneously available, known single CPU bursts on one CPU without switching cost. SRTF preempts according to remaining demand. Unknown bursts require estimates, so the ideal result no longer follows automatically.
- An exchange proof explains the SJF bound. For adjacent available jobs x>y, running x before y contributes x to their combined waiting; swapping contributes y. Later jobs see the same total x+y. Removing every inversion therefore cannot increase total waiting. Different arrivals or deliberate idling require a different argument.

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6.3.1. FCFS and the convoy effect

- FCFS serves arrival order without preemption. A long first job delays many short jobs and can create a convoy through CPU and I/O resources.
- With simultaneous arrivals and bursts 120,10,10 in that order, waits are 0,120,130. Their mean is 250/3, approximately 83.33 time units.

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6.3.2. SJF and SRTF

- SJF selects the shortest available next burst when the CPU becomes available. SRTF can preempt when a job with shorter remaining demand arrives.
- Continually arriving short jobs can starve a long job. An arbitrary tie-break is not a fairness guarantee.

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6.3.3. Round Robin

- RR rotates a ready queue using quantum q. A quantum large enough to finish every burst approaches FCFS; very small q increases timer and possible switching overhead.
- For arrivals zero, bursts 24/3/3, and q=4, the timeline is P1[0,4), P2[4,7), P3[7,10), P1[10,14), P1[14,18), P1[18,22), P1[22,26), P1[26,30). The total is 30. Once only P1 remains, a quantum boundary need not switch to a different thread.

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6.3.4. Priorities and multilevel queues

- Priority policies can starve lower-priority work. Aging increases effective priority with waiting, but its rate, saturation, ties, and arrival assumptions determine whether waiting is bounded.
- Multilevel queues separate classes such as foreground and background. Feedback queues allow class migration according to behavior; quanta and promotion rules are part of the policy.

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6.3. Model, ordering, and boundary cases

Model assumptions. Arrivals, actual or estimated bursts, preemption, and ties are specified.

Invariant and ordering. Timeline intervals do not overlap. Each execution segment consumes no more than remaining demand, and arrivals and completions update the ready set.

Counterexample or boundary. A stream of underestimated short jobs can postpone a long job indefinitely.

Worked check. Simultaneous bursts 8/4/2 give FCFS waits 0/8/12, mean 20/3, and SJF execution order P3/P2/P1 with waits 0/2/6 in execution order, mean 8/3. RR q=2 gives completions P1/P2/P3=14/10/6; total useful execution remains 14.

Reasoning prompt. Which optimality statement must change if the burst ordering uses inaccurate predictions?

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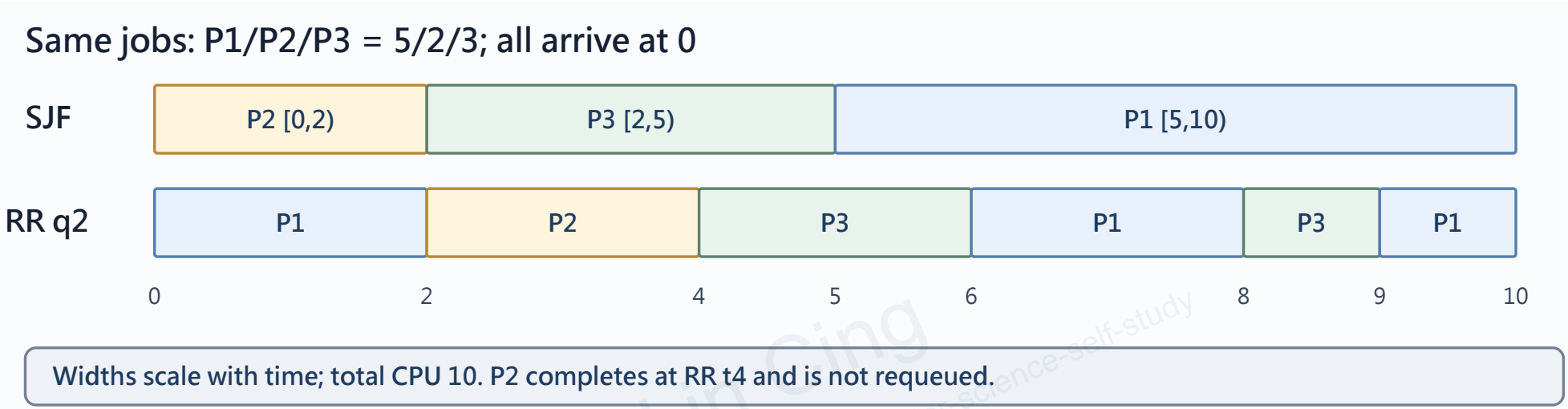
SJF and RR on the same 5/2/3 input

P1/P2/P3 all arrive at 0, bursts are 5/2/3, and switch cost is zero. SJF runs P2[0,2), P3[2,5), P1[5,10). In process order, completions are 10/2/5 and waits 5/0/2.

RR q=2 starts queue P1, P2, P3. It runs P1[0,2), P2[2,4), P3[4,6), P1[6,8), P3[8,9), P1[9,10). P2 completes at 4 and is never requeued. Completions are 10/4/9, waits 5/2/6, and first responses 0/2/4. Both consume 10 useful units; mean waits are 7/3 and 13/3.

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SJF and RR on the same 5/2/3 input (continued)



The separate 8/4/2 case instead has RR q=2 segments P1[0,2), P2[2,4), P3[4,6), P1[6,8), P2[8,10), P1[10,12), P1[12,14), and completions 14/10/6. Mixing those inputs changes the problem.

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6.4. Thread scheduling

- A library can schedule user threads within process-contention scope. Kernel-visible entities compete in system-contention scope.
- One-to-one mapping normally exposes each thread to kernel scheduling. In many-to-one, a genuinely blocking kernel call can stall the process's other user threads unless the runtime avoids that blocking.

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6.4. Model, ordering, and boundary cases

Model assumptions. The mapping and the runtime's handling of blocking calls are known.

Invariant and ordering. One kernel entity runs at most one context at a time. A runtime cannot promise independent progress without an available execution carrier.

Counterexample or boundary. The sole carrier blocking on I/O stops all other runnable tasks in the many-to-one example.

Worked check. T1 awaits network, T2 needs 4 CPU units. With one-to-one mapping T2 can run [0,4) while T1 blocks. With one carrier and a genuinely blocking T1 syscall, T2 has no guaranteed start, so completion at 4 is unjustified.

Reasoning prompt. Which mapping and blocking assumptions support a thread API's concurrency promise?

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6.5. Multiprocessor scheduling

- Affinity can preserve cache and TLB locality. Excessive binding can leave CPUs idle while one queue is overloaded, so balancing trades locality against parallel use.
- Shared queues, migration, and work stealing must prevent a runnable thread from being dispatched on two CPUs simultaneously.

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6.5. Model, ordering, and boundary cases

Model assumptions. Queue ownership and migration synchronization are defined, with cache and communication costs.

Invariant and ordering. A thread has only one actual running owner. Balancing respects required affinity constraints.

Counterexample or boundary. Binding every task to one core preserves some locality while wasting other cores.

Worked check. Fix eight cores, all jobs ready at t=0, indivisible jobs, and no job running on two cores at once. A/B each need 8 ms, C/D each need 4 ms, and E/F/G/H/I/J each need 2 ms: 36 ms of useful work. CPU0 initially queues [A,I,J]; CPU1 through CPU7 initially hold B,C,D,E,F,G,H respectively.

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6.5. Model, ordering, and boundary cases (continued)

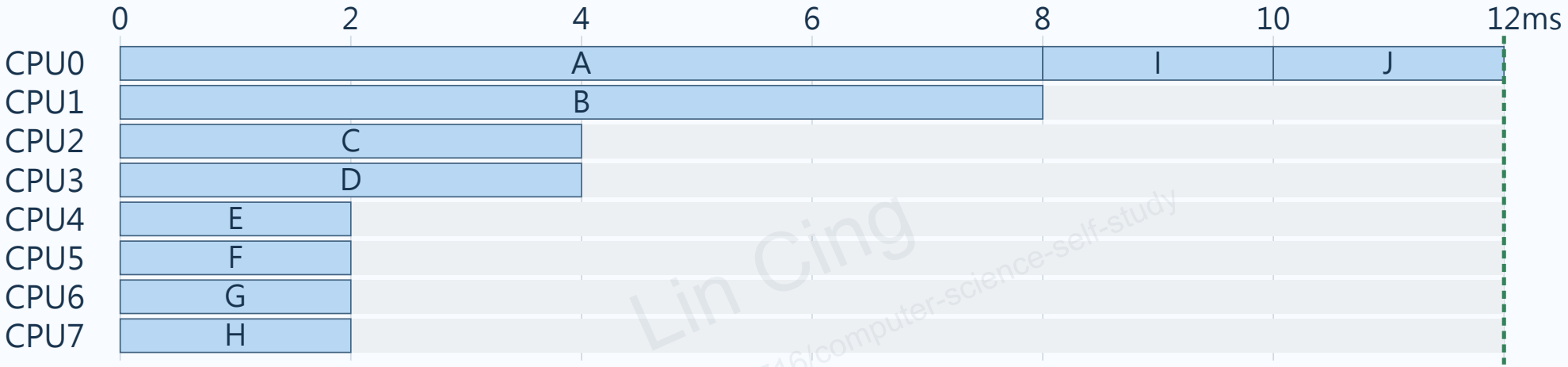
Without migration CPU0 runs A[0,8), I[8,10), J[10,12), so makespan is 12 ms. At t=2 CPU4/5 finish E/F. Under queue synchronization they each remove one still-queued job I/J and transfer its unique ownership. Charge each migration 0.5 ms for queue/transfer and 0.5 ms for cache warmup. Both destination cores pay overhead [2,3), then run I/J [3,5). CPU0 finishes A at 8; makespan becomes 8 ms, speedup=12/8=1.5. Useful work remains 36 ms; added CPU work is 2 ms. These are specified model costs, not hardware measurements.

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6.5. Model, ordering, and boundary cases (continued)

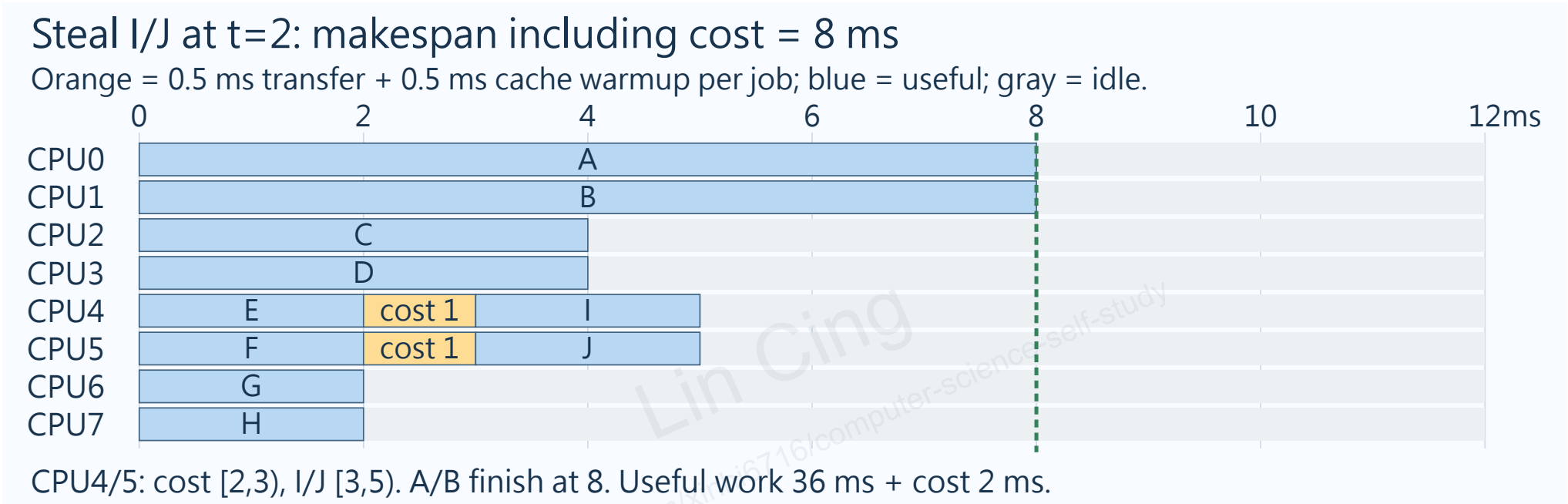
No migration: CPU0 runs A, I, J; makespan = 12 ms

Blue = useful work; gray = idle. All jobs ready at t=0. Identical 0–12 ms axes.



At t=2 CPU4/5 finish E/F, but I/J still wait in CPU0 queue behind A.

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Indivisible A gives the lower bound $T \geq 8$, reached here. With per-migration cost c instead, makespan=max(8,2+c+2)=max(8,4+c). Strict improvement over 12 ms requires $c<8$. Dependencies, mandatory affinity, or different cache costs change the model; idle cores alone do not prove migration beneficial.

6.6. Real-time CPU scheduling

- Real-time requirements concern deadlines. Soft real-time permits defined deadline misses; hard real-time requires a guarantee under its specified workload and failure model.
- Describe a periodic task by execution upper bound e_i, period p_i, and relative deadline d_i. U=sum(e_i/p_i) is load. For independent, preemptible tasks on one ideal CPU, d_i=p_i, and no blocking or overhead, RM has sufficient bound U<=n(2^(1/n)-1), while EDF is schedulable exactly when U<=1 in this model.
- With e1=2, p1=d1=5 and e2=3, p2=d2=10, U=0.7 is below 2(sqrt(2)-1), approximately 0.828. RM gives T1[0,2), T2[2,5), T1[5,7), idle[7,10). Both tasks meet deadlines in the hyperperiod, repeated under the stated assumptions.

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6.6.1. RM, EDF, and resource priorities

- RM assigns shorter periods higher fixed priority. Its classical sufficient bound relies on independent tasks with deadlines equal to periods.
- EDF chooses the earliest absolute deadline dynamically. Its ideal one-CPU utilization result does not specify graceful behavior under overload.
- Priority inheritance temporarily raises a blocking owner's priority. Ceiling protocols provide different restrictions and bounds. Neither replaces analysis of execution and blocking costs.

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6.6. Model, ordering, and boundary cases

Model assumptions. Release patterns, deadlines, execution upper bounds, processor count, switching, and blocking are defined.

Invariant and ordering. Each job completes by its absolute deadline. A blocking bound must be justified; merely enabling priority inheritance does not prove schedulability.

Counterexample or boundary. U<=1 alone does not prove arbitrary hard-real-time workloads meet deadlines when overhead or blocking is omitted.

Worked check. The two-task 0.7 example is schedulable in its ideal model. Add bounded overhead and resource blocking before transferring that conclusion to an implementation.

Reasoning prompt. Which violated assumption prevents utilization alone from establishing a deadline guarantee?

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Historical scheduler mechanisms

Solaris converts class priorities to global priorities. TS/IA uses feedback; priority 0 has a 200 ms quantum, priority 59 a 20 ms quantum. CPU exhaustion and sleep return use different new priorities. Windows 7 uses 32 ready queues with versioned boosts. These are historical mechanisms rather than present-day specifications.

Linux 3.2 CFS stores ready entities in a red-black tree keyed by vruntime; current is outside the tree. The increment is actual execution $\times$ 1024/weight. Nice 0 has weight 1024; nice 5 has 335. Select the smallest vruntime, subject to the version's bounded buddy preferences.

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Historical scheduler mechanisms (continued)

Same workload, three ready structures

Solaris TS/IA

global priority

p20 → q120 ms

sleep return → p50

highest global priority

equal priority: round robin

IA favors window apps

Windows 7

32 ready queues

p1–15 variable / p16–31 RT

time quantum expires → lower

I/O / foreground → boost

foreground quantum ≈ 3×

scan highest nonempty queue

Linux 3.2 CFS

red-black tree

key = vruntime

nice 0/5 weight 1024/335

2 runnable → 3 ms slice

tick: delta_exec > slice

HRTICK queues exact slice

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A fixed Linux 3.2 event model

One CPU, P1 burst 8 ms, P2 burst 4 ms, both runnable at 0. P3 burst 2 ms sleeps until 9. All have nice=0, weight=1024, and normalized initial vruntime=0. P1 enters the tree first on equal keys. Fix GENTLE_FAIR_SLEEPERS=true, wakeup granularity=1 ms, CONFIG_SCHED_HRTICK=y. Switch cost, interrupt cost, and timer jitter are zero. Normalization does not claim a real boot-time START_DEBIT offset; one common shift preserves relative comparisons.

For n runnable entities,

$$period = \begin{cases} 6\ \text{ms}, & n \le 8, \\ 0.75n\ \text{ms}, & n > 8, \end{cases} \qquad ideal_i = period \frac{weight_i}{\sum_j weight_j}.$$

Ideal runtime is a wall-time slice, not a global quantum. Two equal tasks get 3 ms each; three get 2 ms each. Current leaves the ready tree and returns with updated vruntime on preemption; blocked/completed tasks are not ready. HRTICK, a satisfied wakeup condition, or completion must justify a switch. No event at t=8 justifies splitting P1's [6,9) slice.

CFS event trace

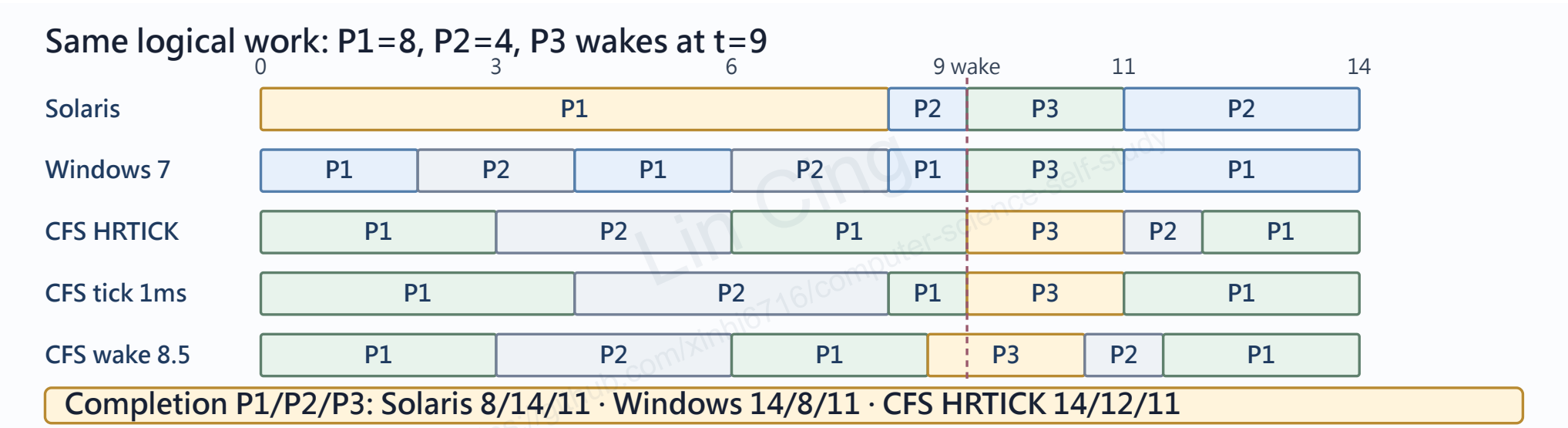
States below are after the event and next-current selection. Ready excludes current; min_vruntime never moves backward.

Time/event	Current	Other ready	min_vruntime	New ideal	Reason
0 initial	P1	P2@0	0	3	Equal keys, P1 inserted first
3 HRTICK	P2	P1@3	0	3	P1 used slice
6 HRTICK	P1	P2@3	3	3	Equal keys, P1 already leftmost
9 HRTICK/wake	P3	P2@3, P1@6	3	2	P3 placed at 0, lead 6>1
11 P3 complete	P2	P1@6	3	3	P2 key 3 precedes 6
12 P2 complete	P1	None	6	6	Only P1, remaining 2
14 P1 complete	None	None	8	None	No work remains

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CFS event trace (continued)

P3 placement at 9 is $\max(v_{sleep}, min_vruntime - 3) = \max(0, 3 - 3) = 0$. P1's key is 6. Wakeup compares current−wakeup>1 ms, so 6−0>1 and P3 immediately runs at 9, not earlier. Gantt is P1[0,3), P2[3,6), P1[6,9), P3[9,11), P2[11,12), P1[12,14). Completions are 14/12/11, not 14/9/11.



Tick, HRTICK, and wakeup sensitivity

Settings	Gantt	Completion P1/P2/P3
HRTICK, P3 wake 9	P1 0–3, P2 3–6, P1 6–9, P3 9–11, P2 11–12, P1 12–14	14/12/11
No HRTICK, 1 ms tick, wake 9	P1 0–4, P2 4–8, P1 8–9, P3 9–11, P1 11–14	14/8/11
HRTICK, wake 8.5	P1 0–3, P2 3–6, P1 6–8.5, P3 8.5–10.5, P2 10.5–11.5, P1 11.5–14	14/11.5/10.5

check_preempt_tick uses strict delta_exec>ideal_runtime. At t=3, 3>3 is false; the ordinary tick first preempts at t=4. Queued HRTICK instead directly requests rescheduling at the programmed slice boundary, without reapplying that strict comparison. Without HRTICK, P2 finishes at 8. At t=9 min_vruntime=5, so P3 returns at max(0,5−3)=2; P1 is 5, and 5−2>1 gives the wake preemption.

Tick, HRTICK, and wakeup sensitivity (continued)

The periodic tick also checks, after minimum granularity, whether current leads the tree's leftmost by more than ideal runtime. This handles fairness discrepancies but does not replace wakeup comparison. Last/next/skip buddy are locality/scheduling hints subject to virtual-time fairness. In this example n<8 before wake, so there is no ordinary next buddy from the sleeping P3; a successful wakeup comparison can nominate it. A buddy too far ahead cannot override the fairness bound.

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Nice weights and historical comparisons

With only A nice 0 and B nice 5 runnable, total weight=1359 and period=6 ms.

Task	Weight	Ideal runtime	vruntime per real ms	CPU share
A	1024	$6\times1024/1359\approx4.521$ ms	1 ms	75.35%
B	335	$6\times335/1359\approx1.479$ ms	$1024/335\approx3.057$ ms	24.65%

Weights are not linear nice percentages. Above eight runnable tasks the period grows to preserve minimum granularity instead of shrinking slices without bound.

Fixed case	Rules	Logical workload completion P1/P2/P3
Solaris TS/IA	p20/q120 ms, sleep return p50	8/14/11
Windows 7	32 queues, normalized 2/6 ticks, I/O boost	14/8/11
Linux 3.2 CFS	HRTICK, nice 0, 6 ms period, P3 wake 9	14/12/11

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Nice weights and historical comparisons (continued)

These are specified event models, not product benchmarks. The same workload alone does not determine one Gantt. Timer mode, wake time, insertion ties, and placement are inputs; changing them yields the alternate completions above.

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6.8. Algorithm evaluation

- Deterministic models calculate a trace exactly. Queueing models use arrival and service distributions. Simulation supports more complex policies. An implementation captures actual hardware and OS costs.
- Disclose workload distribution, processors, switching, and observation window. Means can hide starvation and long tails.

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6.8. Model, ordering, and boundary cases

Model assumptions. Methods use comparable inputs and costs. Random simulations retain seeds and an appropriate uncertainty analysis.

Invariant and ordering. If models disagree on the same simplified workload, investigate omitted costs and distribution assumptions. Agreement is not itself an algorithmic invariant.

Counterexample or boundary. One deterministic SJF success does not establish superiority under every burst-estimation error distribution.

Worked check. Use an exactly replayable deterministic comparison: one CPU; P1/P2/P3 arrive at 0 with bursts 5/2/3 ms; no I/O or switching cost. Response means time to first CPU. Define p99 by nearest rank: with three observations, rank ceil(0.99×3)=3 is this batch maximum. Waiting=completion−burst.

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6.8. Model, ordering, and boundary cases (continued)

Policy	Completion P1/P2/P3	Waiting P1/P2/P3	Mean waiting	First response P1/P2/P3	Batch p99 response
FCFS, initial P1/P2/P3	5/7/10	0/5/7	4	0/5/7	7
SJF, P2/P3/P1	10/2/5	5/0/2	7/3	5/0/2	5
RR q=2, initial P1/P2/P3	10/4/9	5/2/6	13/3	0/2/4	4

All use window [0,10], so throughput=3/10 jobs/ms. SJF has smaller mean waiting while RR has smaller batch first-response tail. Neither metric substitutes for the other. These are exact values for three specified jobs, not random simulation or population-p99 estimates; no 95% confidence interval is claimed. Random experiments require arrival/service distributions, seeds, independent replication units, and an estimator. A confidence interval for a mean is not a range containing 95% of individual waits.

6.8. Model, ordering, and boundary cases (continued)

Reasoning prompt. How would an evaluation expose the tail rather than only mean throughput?

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6.9. Summary

- Scheduling correctness includes queue ownership and no double execution. Service quality adds explicitly required waiting bounds and response distributions.
- High CPU utilization does not imply short user delay. Additional runnable work can increase shared-resource contention and reduce useful throughput.

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6.9. Model, ordering, and boundary cases

Model assumptions. Correctness, quality, and cost receive separate evidence.

Invariant and ordering. If starvation freedom is required, establish it under admission and arrival assumptions. Fixed priorities alone may not provide it.

Counterexample or boundary. A 99% busy batch CPU that indefinitely postpones interactive work fails an interactive-service requirement.

Worked check. Retain RR q=2, all arrivals 0, and initial order P1/P2/P3; multiply every burst by k. Pop the queue head, run min(2,remaining), and append it only if unfinished. Exact replay gives:

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6.9. Model, ordering, and boundary cases (continued)

k; bursts	Completion P1/P2/P3	Waiting P1/P2/P3	Throughput jobs/ms	Batch p99 first-CPU response	Maximum waiting
1; 5/2/3	10/4/9	5/2/6	3/10	4	6
2; 10/4/6	20/10/16	10/6/10	3/20	4	10
3; 15/6/9	30/16/25	15/10/16	1/10	4	16

For k=2 the complete trace is P1[0,2), P2[2,4), P3[4,6), P1[6,8), P2[8,10), P3[10,12), P1[12,14), P3[14,16), P1[16,18), P1[18,20). First responses remain 0/2/4 and busy utilization remains 100%, yet maximum ready waiting grows 6→10→16. First-CPU response, throughput, and waiting measure different effects. A finite batch does not prove starvation freedom under unbounded arrivals.

Reasoning prompt. Which per-task fairness measure reveals starvation hidden by aggregate utilization?

Chapter conclusions

- Scheduling names require arrival and burst assumptions
- Response and turnaround are different measures
- Multiprocessor and real-time analysis must specify locality, deadlines, and failure assumptions

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Exercises and sources

- the separate exercises PDF.
- the separate solutions PDF.
- Coverage and source limitations.

Reference: A. Silberschatz, P. B. Galvin, and G. Gagne, *Operating System Concepts*, 9th edition, Wiley, 2013. These are original explanations and original exercises. Historical product cases refer to the historical versions in the edition. No publisher solution text is used.

- Linux v3.2 sched_fair. c defines period/slice, sleep placement, tick, HRTICK, and wakeup comparisons.
- Linux v3.2 sched. c weight table.
- Microsoft priority boosts and Solaris time-sharing class.

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References and attribution

Abraham Silberschatz, Peter B. Galvin, Greg Gagne. Operating System Concepts, 9th Edition.

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